

PROFESSIONAL SUMMARY

Systems Engineering undergraduate experienced in algorithmic design, object-oriented software development, and embedded control systems. Focused on building scalable logic, ensuring system reliability, and delivering seamless integration between software and hardware environments.

TECHNICAL SKILLS

- **Languages:** C, C++, Java, MATLAB, Julia, CLINGO
- **Embedded & Hardware:** Arduino, Cypress PSoC, Sensors & Actuators
- **Systems & Automation:** Control Theory (PID, Bode/Nyquist)
- **Software & Algorithms:** OOP, Data Structures (Trees, Heaps, Hash Tables), Graph Algorithms (DFS/BFS), Dynamic Programming
- **Tools & Environment:** Git, GitHub, Linux/Bash, Arduino IDE, PSoC Creator, MATLAB, AutoCAD, Netica

ENGINEERING & SOFTWARE PROJECTS

Algorithmic Data Pipeline ("The Royal Hunt") | C, Algorithms & Data Structures

- Architected a multi-stage data processing pipeline in C, intelligently chaining Queues and Binary Search Trees (BST) to parse, filter, and sort raw data extracted from CSV files.
- Optimized data retrieval and dynamic ranking by implementing Hash Tables for $O(1)$ record lookups and Max Heaps to efficiently update attributes and extract top-performing entities.
- Implemented Depth-First Search (DFS) graph traversal algorithms to calculate and evaluate all valid network routes for automated entity assignment based on dynamic criteria.

Automated Smart Home Safety & Exhaust System | C++, Arduino, OOP

- Engineered an automated safety system using an Arduino Uno to monitor hazardous gas levels (MQ-7) and trigger a DC fan and servo-driven window.
- Structured the C++ firmware into modular OOP classes, utilizing a time-based `millis()` state machine to enable concurrent sensor reading and smooth, incremental servo actuation.
- Implemented an autonomous recovery cycle that continuously evaluates air quality, resetting actuators once gas concentration drops below safe thresholds.

Distributed Smart Irrigation System | Embedded C++, SPI, ISRs

- Developed a distributed IoT irrigation system using two Arduino microcontrollers communicating via Master/Slave SPI, where the Master processes environmental data (Bluetooth, DHT11) to control the Slave.
- Implemented hardware timers and Interrupt Service Routines (ISRs) for real-time PIR motion detection, instantly overriding actuators for fail-safe operation.
- Structured the embedded C++ codebase using Object-Oriented principles, encapsulating hardware states into reusable, modular classes.

Enterprise Cloud Calendar System | Java, OOP, Data Structures

- Developed a cloud-based calendar scheduling application in Java (similar to Google Calendar), implementing a object-oriented architecture to manage overlapping events, fixed-duration meetings, and participant responses.
- Implemented a multi-tier account system (Staff, Managers, Guests) that enforces specific business rules—such as priority-based scheduling limits, automatic high-priority invitation acceptance, and conflict overrides.
- Designed a modular codebase using object-oriented principles to handle event identification, multi-user invitations, and real-time status tracking (accepted, declined, pending).

AI & Predictive Modeling Models | Netica, Clingo, A*, Simulated Annealing

- Implemented optimization & search algorithms (Simulated Annealing, A*, MinMax), Logic Programming, and Bayesian Networks for predictive modeling.

EDUCATION

Faculty of Automatic Control and Computers , University POLITEHNICA of Bucharest | Expected 2028

B.Sc. Systems Engineering (*Automation and applied informatics*)

- *Pedagogical Training Module (Ongoing)*

Erasmus+ Mobility Program | NOVA School of Science and Technology | September 2025 - January 2026

Mihai Viteazul National College, Bucharest

- Baccalaureate: Mathematics (10), Informatics (9.95), Romanian (9.65).